

Adiba Yaseen Research and Teaching Portfolio

Artificial Intelligence and Machine Learning
Computational Biology and Bioinformatics

Research portfolio for university faculty and AI research opportunities in Riyadh and Saudi Arabia

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PhD in Computer Science Final CGPA 3.92 out of 4.00

Based in Riyadh Saudi Arabia

Professional Positioning

I am a computer scientist, AI and machine learning researcher, and university lecturer based in Riyadh. My work focuses on computational biology and AI driven drug discovery, with particular emphasis on protein protein and compound protein interactions, therapeutic peptide safety, molecular representation learning, graph neural networks, and realistic performance evaluation. I combine research software, curated biological datasets, reproducible experiments, and university teaching to translate technical methods into work that academic teams, students, and interdisciplinary collaborators can use.

My academic preparation includes a PhD in Computer Science completed in December 2024 with a final CGPA of 3.92 out of 4.00, an MS in Computer Science with a CGPA of 3.62, and a BS in Computer Engineering with a CGPA of 3.351. I have nearly five years of teaching experience across university and college appointments, including Artificial Intelligence, Machine Learning, programming, object oriented programming, databases, operating systems, networks, and introductory information technology.

Role Alignment

- University Lecturer and Assistant Professor roles in computer science, artificial intelligence, data science, bioinformatics, and related programs
- AI and machine learning research roles involving rigorous experimentation, scientific computing, and model evaluation
- Computational biology, bioinformatics, molecular AI, and AI driven drug discovery research
- Interdisciplinary teaching and research programs connecting computer science with healthcare, biotechnology, and life sciences

Research Evidence Overview

Research stream	Principal output	Status	Documented evidence
Therapeutic peptides	HemoNet	Published 2021	88 percent AUC ROC; external, nonredundant and clinical peptide validation
Compound protein interaction	Kernel CPI and HKRCPI	Published 2022	RFPP screening metric; about 908250 protein drug pairs scored
Protein complex inhibition	PPI Inhibitors	Archived preprint 2024	GNN model; 0.85 cross validation AUC ROC and two external evaluations
Protein binding affinity	MS thesis and IBCAST paper	Published 2018	SVR with interfacial features using 135 three dimensional protein structures

Research Approach

My research trajectory moves from structural protein analysis and support vector regression to contextual peptide embeddings, compound protein screening, and graph neural networks for targeted protein complex inhibition. Across these projects, a consistent priority is evaluation that reflects real use. I have worked with similarity controlled splits, nonredundant cross validation, independently collected external data, experimentally verified negatives, and ranking measures designed for large screening libraries.

The public research repositories connect publications with code, datasets, trained models, notebooks, supplementary results, and prediction workflows. This makes the work suitable for collaboration, student research supervision, reproducibility exercises, and further development in academic or applied AI settings.

Research Case Study HemoNet

Research question How can a computational model predict peptide hemolytic activity while accounting for amino acid context, L and D amino acids, and chemical modifications at peptide termini?

Scientific relevance Hemolytic activity is a key safety concern in therapeutic peptide development. Computational screening can help prioritize candidates before costly laboratory testing.

Data The study filtered DBAASP version 2 and Hemolytik records and formed a dataset of 2056 hemolytic and 2883 nonhemolytic peptide examples. It also used a compositionally distinct external set and 28 clinical antimicrobial peptides.

Representation HemoNet used contextual SeqVec embeddings for peptide sequences. It represented N and C terminal modifications with SMILES derived extended connectivity fingerprints produced with RDKit, allowing the model to handle previously unseen modification types.

Modeling The primary model was a fully connected PyTorch neural network using transfer learned SeqVec features and terminal modification features. Comparisons included SVM, random forest, XGBoost, BLAST, HemoPI, and HemoPred.

Evaluation The work used repeated stratified five fold cross validation, 90 percent and 70 percent sequence identity controlled nonredundant cross validation, independent external evaluation, and assessment on clinical peptides.

Results HemoNet achieved about 88 percent AUC ROC in the principal comparison, versus about 73 percent for HemoPI. Performance decreased in the expected direction under stricter nonredundancy, from about 88 percent to 81 percent and 77 percent. On the independent external set, the 70 percent nonredundant model achieved 72 percent AUC ROC, compared with 40 percent for HemoPI and 44 percent for HemoPred. Almost all clinical peptides received low predicted hemolysis scores; the exception was Nisin, for which laboratory hemolytic response had been reported.

Research contribution The project integrated biological sequence context with chemical representations of terminal modifications, emphasized strict generalization testing, and released the implementation and supplementary resources for reuse.

Portfolio evidence Published journal article, GitHub repository, datasets, trained model resources, repeated evaluation scripts, clustering code, UMAP visualizations, and external validation data.

Publication doi.org/10.1142/S0219720021500219

Code and data github.com/adibayaseen/HemoNet

Research Case Study: Compound Protein Interactions

Research question How should machine learning models for compound protein interactions be evaluated so that reported performance reflects generalization and target compound screening use?

Problem identified Standard random splits can place similar or duplicate protein compound examples in training and test sets. Synthetic negative construction and general classification metrics can also create overoptimistic estimates that do not match large library screening.

Method The study compared CPI NN, GraphDTA, and a simpler Kernel CPI model. It controlled similarity between training and test examples, examined random and similarity controlled negative generation, tested experimentally verified negatives from BindingDB, and used nonredundant cross validation.

Screening metric RFPP measures the rank of the first known positive compound for each protein. It translates model performance into the number of candidates that may need to be screened before finding a known interaction.

Results Under nonredundant cross validation, Kernel CPI achieved 69.98 percent AUC ROC and 77.30 percent AUC PR. In an all versus all experiment covering about 292500 combinations, the model placed at least one known interacting compound in the top 10 for 85 percent of test proteins.

Drug repurposing study The model scored about 908250 pairs formed from NR HCPI proteins and 3633 FDA approved drugs in SuperDRUG2. It placed at least one known interaction in the top 10 for 75 percent of proteins, compared with 50 percent for CPI NN and 3 percent for a random predictor.

Biomedical application The repository includes top ranked and supporting results for human ACE2 and SARS CoV 2 Spike screening. These outputs are computational candidates rather than clinical conclusions.

Research contribution The work joins experimental design, interpretable evaluation, scalable screening, and reproducible software. It demonstrates the ability to question misleading benchmark assumptions and build an evaluation protocol around the intended scientific use.

Publication doi.org/10.1093/bioinformatics/btac496

Code and supplementary results github.com/adibayaseen/HKRCPI

Research Case Study Targeted Protein Complex Inhibition

Publication status Archived bioRxiv preprint from 2024. This work is presented as a preprint and has not been described as peer reviewed.

Research question Can a model take a protein complex and a compound as inputs and predict whether the compound will inhibit the interaction between the proteins in that complex?

Model The project used a graph neural network representation of the protein complex together with a molecular representation derived from the compound SMILES string. It compared the GNN pipeline with SVM and GearNet based alternatives.

Data engineering The repository integrates protein complex and inhibitor information from 2P2I, negative protein complexes from DBD5, random compound negatives from SuperDRUG2, molecular similarity filtering, protein sequence and structural features, and trained model artifacts.

Evaluation The model was trained and validated on 714 inhibitors of 23 protein complexes. It reported cross validation AUC ROC of 0.85, an external literature dataset AUC ROC of 0.82, and AUC ROC of 0.78 on putative inhibitors of the SARS CoV 2 Spike and human ACE2 complex.

Research contribution The project formulates targeted protein protein interaction inhibitor prediction as a paired protein complex and compound learning problem, develops an end to end graph based pipeline, and provides reconstructable notebooks and data resources.

Portfolio evidence Archived preprint, GitHub repository, data preparation files, GNN notebooks, SVM baselines, GearNet comparisons, trained models, and result folders.

Archived preprint doi.org/10.1101/2024.08.23.609286

Code and data github.com/adibayaseen/PPI-Inhibitors

Research Case Study Protein Binding Affinity

Research question Can structural and interfacial features support accurate protein protein binding affinity prediction while remaining useful when protein conformations change?

MS thesis The thesis used 135 three dimensional protein structures and addressed interaction prediction, binding affinity prediction, and analysis of the influence of structure based features.

Methods Supervised learning, classification, SVM, SVR, Python based analysis, and protein interface features.

Research development The work was presented at ISACIS 2017 and developed into a peer reviewed IBCAST conference paper on support vector regression with interfacial features.

Research value This project established the structural bioinformatics and classical machine learning foundation for later work in sequence representations, compound protein interactions, and graph based molecular learning.

Conference paper doi.org/10.1109/IBCAST.2018.8312222

Collaborative Study Sequence Based Affinity Prediction

I coauthored the ISLAND study, which examined protein binding affinity prediction from sequence information when three dimensional structures are unavailable. The paper evaluated generalization against an independent test set, reported limitations in existing sequence only predictors, and introduced an improved sequence based predictor. The study was published in BioData Mining in 2020.

Journal article doi.org/10.1186/s13040-020-00231-w

Open Source and Reproducible Research

The public repositories are research records rather than isolated software demonstrations. Together they contain preprocessing code, curated examples, prediction scripts, experimental notebooks, trained model links, result files, supplementary documents, and workflows that connect each input to an interpretable output.

HemoNet Repository

- Hemolytic and nonhemolytic peptide datasets, external data, and 28 clinical antimicrobial peptide records
- Five fold and nonredundant cross validation code, repeated experiment aggregation, BLAST baselines, CD HIT based clustering, and UMAP visualization
- Prediction workflow for peptide sequence and terminal modification inputs

HKRCPI Repository

- NR HCPI data, BindingDB derived negative examples, SuperDRUG2 compounds, and comparative kernel experiments
- RFPP calculation, all versus all target compound screening, pair prediction, and drug repurposing workflows
- ACE2 and Spike top predictions, median rank results, and supporting evidence files

PPI Inhibitors Repository

- Protein complex and inhibitor data from 2P2I, DBD5 negatives, SuperDRUG2 compounds, and similarity aware binders
- Graph neural network training, SVM baselines, GearNet embeddings, engineered protein complex features, trained model directories, and final results
- Reconstructable notebooks that link the archived preprint to data and model evaluation

GitHub profile github.com/adibayaseen

Technical Capability

Programming and environments Python, Jupyter Notebook, C plus plus, C, C sharp, SQL, MATLAB, PyTorch, TensorFlow, RDKit, BioPython, and PyMOL.

Machine learning Neural networks, graph neural networks, CNNs, SVM and SVR, random forests, XGBoost, kernel methods, transfer learning, classification, regression, representation learning, feature engineering, and model comparison.

Computational biology Protein sequence and structure analysis, protein protein interactions, compound protein interactions, binding affinity, peptide activity and toxicity, molecular representations, SMILES, fingerprints, molecular similarity, and target compound screening.

Research evaluation Stratified and nonredundant cross validation, independent external evaluation, similarity controlled data partitions, negative example construction, AUC ROC, AUC PR, MCC, sensitivity, specificity, screening ranks, and reproducibility.

Teaching Portfolio

My teaching experience covers foundational computing and advanced AI topics. I aim to make technical concepts understandable through worked examples, live programming, laboratory practice, structured assessment, and guided feedback. My research background supports project based teaching in which students learn how to formulate a problem, prepare data, select methods, evaluate results, and communicate limitations.

University of Lahore

November 2020 to July 2024 Lecturer in the Department of Computer Science and Information Technology.

- Artificial Intelligence
- Machine Learning
- Computer Programming
- Introduction to Information Technology
- Database Management Systems
- Operating Systems and Networks

September 2017 to June 2018 Lecturer in the Department of Computer Science and Information Technology.

- Programming Fundamentals I in C plus plus
- Object Oriented Programming in C plus plus

Government Degree College for Women Shahkot

October 2015 to February 2016 Lecturer in Computer Science for intermediate level classes using the Punjab curriculum.

Courses for Future Contribution

In addition to previously taught courses, my doctoral research and graduate coursework prepare me to contribute to Deep Learning, Pattern Recognition, Bioinformatics, Computational Biology, Data Mining, Computer Vision, Medical Image Processing, Molecular AI, and graduate research methods. These areas are presented as teaching capability grounded in research and advanced study, not as courses already taught.

Student Research and Supervision Themes

- Machine learning experiments with transparent baselines and generalization focused validation
- Bioinformatics projects using protein sequences, structures, peptides, and molecular representations
- Graph neural networks for molecular and biological prediction
- Reproducible research notebooks, model documentation, and responsible interpretation of biomedical predictions

Education and Academic Preparation

PhD in Computer Science | Pakistan Institute of Engineering and Applied Sciences

June 2018 to December 2024 | CGPA 3.92 out of 4.00

Focus Biomedical applications and machine learning

- Doctoral coursework included Applied Bioinformatics, Machine Learning, Machine Learning in Cyber Security, Computational Biomolecular Design, Computational Interactomics, Information Retrieval and Data Mining, and Quantum Programming. The 2020 result record documents 3.91 through semester seven; the final CGPA of 3.92 is the completed degree result confirmed for this portfolio.

Master of Science in Computer Science | Pakistan Institute of Engineering and Applied Sciences

2016 to 2017 | CGPA 3.62 out of 4.00

Focus Bioinformatics and machine learning

- Completed the degree requirements on 1 November 2017. Coursework covered Algorithms Design, Theory of Computation, Advanced Computer Architecture, Advanced Operating Systems, Communication Skills, Digital Image Processing and Analysis, Artificial Intelligence, Pattern Classification and Recognition, Bioinformatics, Medical Image Processing, Advanced Database Systems, Machine Vision, Machine Learning in Bioinformatics, Data Warehousing, and Deep Neural Networks.

Bachelor of Science in Computer Engineering | University of Engineering and Technology Lahore

2011 to 2015 | CGPA 3.351 out of 4.00

Focus Image processing and computer engineering

- The official transcript confirms degree completion in 2015 and includes programming, object oriented programming, data structures and algorithms, computer organization, architecture, operating systems, software engineering, networks, databases, artificial intelligence, control systems, embedded systems, signal processing, industrial automation, bioinformatics, and digital design.

Scholarships and Research Support

- Higher Education Commission sponsored NRPB funded project support for PhD research from June 2018 to June 2022
- PIEAS IT Endowment Fund Fellowship sponsored by the Higher Education Commission for the MS in Computer Science from 2015 to 2017
- Fully funded Pakistan Engineering Congress scholarship for the BS in Computer Engineering from 2011 to 2015
- Merit scholarship supporting intermediate education at Shahab College

Publications

Peer Reviewed Work

- Yaseen, A., Abbasi, W. A., and Minhas, F. U. A. A. (2018). Protein binding affinity prediction using support vector regression and interfacial features. In 2018 15th International Bhurban Conference on Applied Sciences and Technology (IBCAST), 194-198. DOI 10.1109/IBCAST.2018.8312222. [Link](#)
- Abbasi, W. A., Yaseen, A., Hassan, F. U., Andleeb, S., and Minhas, F. U. A. A. (2020). ISLAND: in-silico proteins binding affinity prediction using sequence information. BioData Mining, 13, article 20. DOI 10.1186/s13040-020-00231-w. [Link](#)
- Yaseen, A., Gull, S., Akhtar, N., Amin, I., and Minhas, F. (2021). HemoNet: Predicting hemolytic activity of peptides with integrated feature learning. Journal of Bioinformatics and Computational Biology, 19(5), article 2150021. DOI 10.1142/S0219720021500219. [Link](#)
- Yaseen, A., Amin, I., Akhter, N., Ben-Hur, A., and Minhas, F. (2022). Insights into performance evaluation of compound-protein interaction prediction methods. Bioinformatics, 38(Supplement 2), ii75-ii81. DOI 10.1093/bioinformatics/btac496. [Link](#)

Archived Preprint

- Yaseen, A., Roy, S., Akhter, N., Ben-Hur, A., and Minhas, F. (2024). Predicting small-molecule inhibition of protein complexes. bioRxiv. DOI 10.1101/2024.08.23.609286. [Link](#) *Archived preprint and not peer reviewed*

Presentations and Workshops

- Presented MS research at the International Symposium on Advances in Computer and Information Science 2017 at PIEAS Islamabad
- Presented the binding affinity paper at the 15th International Bhurban Conference on Applied Sciences and Technology in January 2018
- Resource person for Deep Learning activities at PIEAS in September 2016
- Participant in ISACIS 2016 and 2017, Applications of Computational Biology at NIBGE in July 2017, SLATE online learning management faculty development training in December 2017, and the International Conference on Advancements in Computational Sciences in February 2018

Academic Service and Engagement

- Active member of the Pakistani Students Welfare Organization at UET Lahore
- Active member of the ACM student community and Software Square Computer Science Society
- Participation in faculty development, research workshops, academic conferences, and student focused activities

Selected Supporting Projects

Gesture Recognition Using Kinect

Built a C sharp computer vision system using live video, depth, and skeletal data. The work covered visual effects, motion detection, background removal, gesture recognition, and integration into a virtual augmented Pearl Hunting game.

Intrusion Detection

Explored one class support vector classification combined with deep belief networks and GPU based deep neural networks using TensorFlow.

Microarray Cancer Classification

Designed an ensemble classification concept using correlation coefficients and feature selection for high dimensional microarray data.

Face Understanding

Studied automatic feature representation using convolutional and deep neural networks with GPU computing.

Online Hospital Management System

Developed a database backed hospital management web application using ASP.NET and SQL.

Medical Image Enhancement

Applied high pass, low pass, averaging, median, Gaussian, Sobel, and histogram equalization methods using ImageJ and DICOM software.

Reinforcement Learning and Robot Behavior

Explored convergence and behavioral tradeoffs in a simulated robot setting using reinforcement learning concepts.

Contribution in Riyadh

I am positioned to contribute to computer science and AI teaching, interdisciplinary research, student project supervision, and collaborative work connecting machine learning with health and life science problems. My strongest contribution is the combination of classroom experience, a completed PhD, peer reviewed computational biology research, and public research implementations that can support both teaching and new investigation.

Academic contribution Teach core computer science and AI courses, develop research informed course material, supervise student projects, and contribute to curriculum and departmental activities.

Research contribution Design reproducible machine learning studies, evaluate generalization carefully, develop molecular and protein representations, and support computational screening research.

Collaboration contribution Work across computer science, bioinformatics, biotechnology, healthcare, and data science teams while communicating methods and limitations clearly.